

Lecture 15: Kernel Methods

Caltech CS156 Technical Study Guide

Gemini Notebook

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Executive Summary

Kernel Methods extend Support Vector Machines (SVMs) to operate in high-dimensional or infinite-dimensional feature spaces without explicitly computing the transformations. By replacing inner products in Z -space with a Kernel Function $K(\mathbf{x}, \mathbf{x}')$, SVMs achieve linear separability on complex non-linear boundaries. This guide covers the Kernel Trick, Mercer's Condition, Soft Margin SVMs with slack variables, and Support Vector classification.

1 The Kernel Trick

In Hard Margin SVMs, the dual Quadratic Programming (QP) problem depends *only* on inner products between transformed feature vectors $\Phi(\mathbf{x})$ and $\Phi(\mathbf{x}')$:

$$\mathcal{L}_D(\alpha) = \sum_{n=1}^N \alpha_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \langle \Phi(\mathbf{x}_n), \Phi(\mathbf{x}_m) \rangle \quad (1)$$

A **Kernel Function** $K(\mathbf{x}, \mathbf{x}')$ computes this inner product directly in input space $\mathcal{X}$:

$$K(\mathbf{x}, \mathbf{x}') = \langle \Phi(\mathbf{x}), \Phi(\mathbf{x}') \rangle \quad (2)$$

This allows implicit computation in infinite dimensions at low computational cost $O(d)$.

2 Common Kernel Functions

Kernel Type	Mathematical Formula	Feature Space Dimension
Polynomial Kernel	$K(\mathbf{x}, \mathbf{x}') = (1 + \mathbf{x}^T \mathbf{x}')^Q$	$\binom{d+Q}{Q}$ (Finite)
Gaussian RBF Kernel	$K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \ \mathbf{x} - \mathbf{x}'\ ^2)$	Infinite Dimension (∞)
Sigmoid Kernel	$K(\mathbf{x}, \mathbf{x}') = \tanh(\alpha \mathbf{x}^T \mathbf{x}' + c)$	Neural Net Equivalent

Table 1: Standard valid kernel functions used in SVM classification.

2.1 Mercer's Condition

A function $K(\mathbf{x}, \mathbf{x}')$ is a valid kernel if and only if it is symmetric $K(\mathbf{x}, \mathbf{x}') = K(\mathbf{x}', \mathbf{x})$ and its Gram Matrix $\mathbf{K}_{ij} = K(\mathbf{x}_i, \mathbf{x}_j)$ is **positive semi-definite** for any finite set of points:

$$\sum_{i=1}^N \sum_{j=1}^N c_i c_j K(\mathbf{x}_i, \mathbf{x}_j) \geq 0 \quad \forall \mathbf{c} \in \mathbb{R}^N \quad (3)$$

3 Soft Margin SVM (Handling Non-Separable Data)

When data is noisy or non-linearly separable, strict margin enforcement fails. We introduce **slack variables** $\xi_n \geq 0$ to allow margin violations:

$$y_n(\mathbf{w}^T \Phi(\mathbf{x}_n) + b) \geq 1 - \xi_n, \quad \xi_n \geq 0 \quad (4)$$

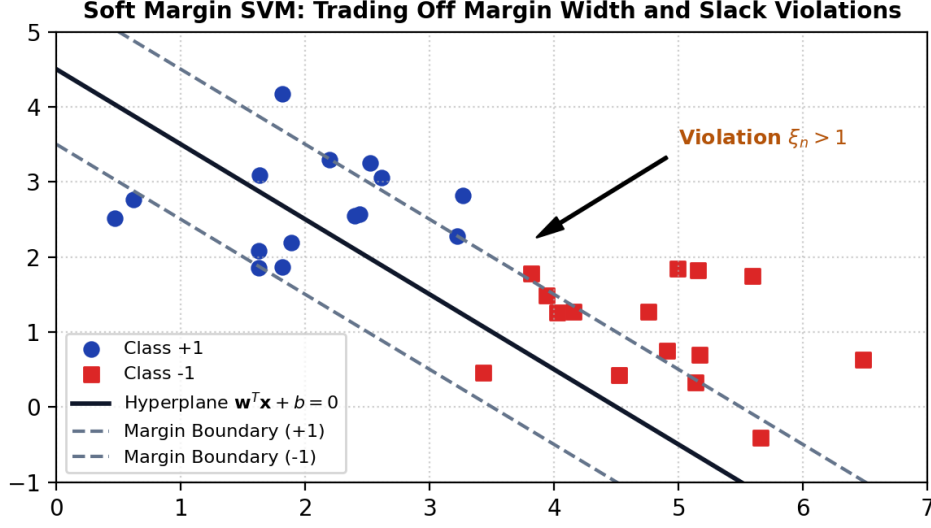


Figure 1: Soft Margin SVM: The parameter C controls the balance between maximizing margin width $\|\mathbf{w}\|^{-1}$ and penalizing margin violations ξ_n .

3.1 Primal and Dual Formulations

The **Primal Objective** balances margin size and total error penalty C :

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N \xi_n \quad (5)$$

The **Dual Objective** replaces the unbounded $\alpha_n \geq 0$ constraint with a box constraint $0 \leq \alpha_n \leq C$:

$$\max_{\alpha} \sum_{n=1}^N \alpha_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m K(\mathbf{x}_n, \mathbf{x}_m) \quad \text{s.t.} \quad 0 \leq \alpha_n \leq C, \sum_{n=1}^N \alpha_n y_n = 0 \quad (6)$$

4 Types of Support Vectors

- **Margin Support Vectors** ($0 < \alpha_n < C$): Sit exactly on the margin boundary ($y_n g(\mathbf{x}_n) = 1, \xi_n = 0$). Used to solve for bias b .
- **Non-Margin Support Vectors** ($\alpha_n = C$): Violate the margin boundary ($\xi_n > 0$). Points with $0 < \xi_n \leq 1$ are correctly classified within the margin; points with $\xi_n > 1$ are misclassified.